LiDAR vegetation structure metrics for Belgium at 30 m

Methods, decisions, checks and results — the Belgian counterpart of the forest covariates of the England wood-ant habitat model

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Repository: github.com/dvanderzande/belgium-lidar-forest-metrics · version 0.1.0 (grid provisional)

Summary

We computed the four LiDAR forest-structure layers of the England wood-ant model (Ant_ENM_England_2025) for all of Belgium: canopy cover, 90th percentile height, height standard deviation and mean height, per 30 m cell from 1 m canopy heights. The definitions are those of the England code; the 1 m input is built from the regional surface and terrain models (DSM − DTM) with buildings removed, encoded like the Environment Agency Vegetation Object Model (VOM) that England used.

- **Delivered:** one GeoTIFF for Belgium (4 bands named as in England, EPSG:31370, 30 m) and a quality layer on the same grid that gives the source survey of every cell and marks a Walloon zone with unreliable values.
- **Sources:** Flanders and Brussels from the Flemish DHMV II (2013–2015); Wallonia from the SPW survey 2013–2014. Both are leaf-off surveys of the same period, 1 m.
- **Two deviations from England** (needed because the Belgian input is DSM − DTM, not a vegetation model): canopy gaps up to 3 m are filled (leaf-off crowns), and heights above 60 m are set to 0 (cranes, turbines).
- **Data-quality zone in Wallonia (code 3):** in Hainaut and Entre-Sambre-et-Meuse (3,240 km², 16 % of the Walloon forest) the 2013–14 survey misses much of the leaf-off broadleaf forest. Cells there should be left out of model fitting, or the zone flagged in predictions.
- **Comparability:** cover and the 0.3 forest threshold behave as in England; absolute heights are higher after gap filling (+3–6 m in forest) and Walloon heights are about 1 m above Flemish ones, so a transfer of the fitted England model needs care.
- **Still open:** the grid, CRS and extent the model needs; transfer of the England model or a new Belgian fit (section 9).

1 Background: the wood-ant model and the England reference

1.1 Goal of the model

Red wood ants (*Formica rufa* group) build large, long-lived mound nests in forests and are declining in many European countries (Balzani et al. 2022). In Belgium the group comprises *F. rufa*, *F. polyctena* and their hybrids (INBO 2008); in north-western Belgium the number of colonies more than halved in 20 years, mainly through deterioration of the habitat rather than its loss (Dekoninck et al. 2010).

The layers in this report are inputs to a species distribution model. Such a model relates known nest locations to environmental layers and predicts habitat suitability at 30 m. The aim is to show where the ants can live, which suitable forest is not occupied, what they can reach by natural spread, and where translocation of colonies could help. Climate, soil and topography describe the wider conditions; the LiDAR layers describe the forest itself.

1.2 Why forest structure matters for wood ants

Nests sit at forest edges, in gaps and along rides. The mound must warm up in the sun, while the trees give shelter and food: honeydew from aphids in the crowns. Shade buffers nest temperature (Chen & Robinson 2014), and in a LiDAR-based habitat model near Basel *F. rufa* and *F. polyctena* differed in canopy cover and distance to the forest edge (Glanzmann et al. 2019). New nests form close to existing ones; England assumes 100 m of natural spread. Each layer measures one side of this (Figure 1): cover is the shade on the ground, p90 the height and age of the stand, SD the variation in canopy height (gaps, edges, mixed tree heights), and cover ≥ 0.3 marks where forest is. The layers come from a surface model, which sees the top of the canopy only: a shrub layer under a closed canopy is not in them.

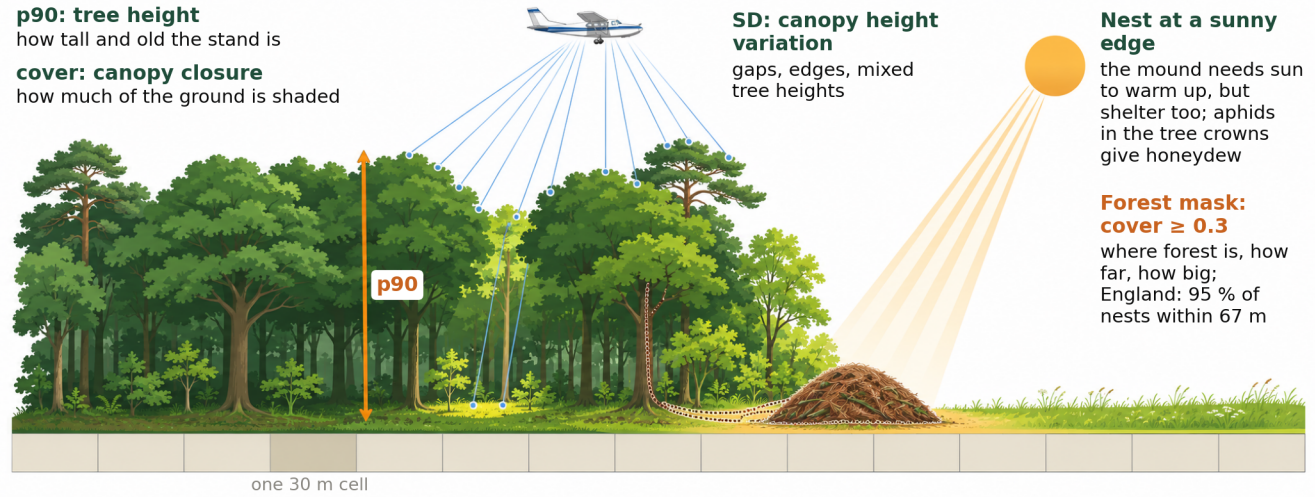


Figure 1. A red wood ant nest at a sunny forest edge, scanned by an airborne LiDAR survey, and what each layer measures. Illustration made with an image generator to our brief; labels added by make_infographics.py.

1.3 The England reference model

The reference is the habitat model for England by Joris Wiethase (repository Ant_ENM_England_2025, github.com/jwiethase/Ant_ENM_England_2025, R). We found no peer-reviewed publication of it (November 2026), so the code is the reference: the definitions of the four metrics in this project follow its scripts exactly (sections 4 and 5).

step	England model
records	*F. rufa* and *F. lugubris* nests from BWARS and field surveys; accuracy ≤ 100 m, spatially thinned
predictors	climate, soil temperature, topography and forest PC1 and PC2; 14 left after a VIF selection
model	Maxent (ENMeval); background points weighted by a recording-effort surface (S7); evaluated with the Boyce index and null models (S9)
outputs	suitability at 30 m for England, also for the 2070–73 climate (forest and topography kept fixed); occupied, suitable and colonisable forest patches and translocation sites (S11)

The LiDAR layers enter the England model in three places:

role	script	how the LiDAR layers are used
model domain	S4b, S6	forest = cover ≥ 0.3, buffered by 70 m (95 % of nests lie within 67 m of forest; 525 of 548 nest records inside); background points and predictions only in this domain; distance to forest was tried and dropped
forest covariates	S5	PCA of cover, √p90 and √SD inside the domain gives forest PC1 and PC2; the PCA centring and scaling are not saved
patches and translocation	S11	forest patches (cover ≥ 0.3); patches smaller than those around 99 % of nest records are not counted as suitable; occupied patches within 50 m of nests; natural spread 100 m; translocation 0, 2, 5 and 25 km away on Forestry England land

Because cover enters as a threshold and as patch areas, and p90 and SD through a PCA, the Belgian layers must be encoded as in England, zeros included (section 4); our two deviations and their effect on comparability are in sections 5 and 8.

1.4 From the layers to Belgian ant populations: a proposal

- 1. Gather nest records with ≤ 100 m accuracy (INBO, Institute of Natural Sciences, volunteer data such as waarnemingen.be), from the LiDAR period 2012–2015 where possible.
- 2. Read the layers at nests and at background points.
- 3. Leave out cells with quality code 3 (the Walloon zone, section 7), or treat them separately.
- 4. Fit a Belgian Maxent model with the England design, or transfer the England model (section 8).
- 5. Validate with exhaustive surveys of known colonies, as England did (Boyce index).

Sources. *Balzani P. et al. 2022, Challenges and a call to action for protecting European red wood ants, Conservation Biology. Chen Y.-H. & Robinson E.J.H. 2014, PLOS ONE (F. lugubris). Dekoninck W., Hendrickx F., Grootaert P. & Maelfait J.-P. 2010, Present conservation status of red wood ants in north-western Belgium, European Journal of Entomology 107: 209–218. Glanzmann L., Perron M. & Freitag A. 2019, Schweizerische Zeitschrift für Forstwesen 170: 32–39. INBO 2008, De rode bosmier in Vlaanderen: voorkomen, bedreigingen en herstelmaatregelen. Wiethase J., Ant_ENM_England_2025 (GitHub repository).*

2 Deliverables

The files are in the project working folder (not in git; large files are kept out of the repository) and can be shared on request.

file	content
belgium/metrics_30m_belgium.tif	4 bands, float32, NoData = NaN (outside Belgium); 9400 × 7600 cells of 30 m; EPSG:31370 (Belgian Lambert 72); cell edges on multiples of 30 m; tiled, deflate-compressed; GeoTIFF tags record sources, rules and the script
belgium/quality_30m_belgium.tif	same grid; uint8 band *source_quality*, NoData = 0; codes below
belgium/quality_zone_wallonia_2013_2014.gpkg	the Walloon data-quality zone as a polygon (EPSG:31370)
flanders/, brussels/, wallonia/ metrics_30m_*.tif	the same metrics per region (identical values inside each region)

band	England name	definition (per 30 m cell, from its 900 pixels of 1 m)
1	cover_VOM	share of pixels with data whose height is > 2 m (0–1)
2	perc09_height_VOM	90th percentile of height (R quantile type 7 = numpy 'linear')
3	sd_height_VOM	standard deviation of height, n − 1 (NaN if fewer than 2 pixels with data)
4	mean_height_VOM	mean height

Pixels without vegetation of at least 2.5 m have height 0 and count in all four metrics, exactly as in the England VOM; NoData pixels (no LiDAR, outside Belgium) are ignored.

code	quality layer meaning	cells (area)
0	no data	–
1	DHMV II (Flanders, Brussels)	15.35 M (13,816 km²)
2	Wallonia 2013–14, reliable	15.18 M (13,660 km²)
3	Wallonia 2013–14, zone where leaf-off broadleaf forest is missed (cover and heights too low)	3.60 M (3,241 km²)

3 Input data

region	heights (1 m)	buildings	border	volume
Flanders	DHMV II DSM and DTM, Digitaal Vlaanderen; flown 2013–2015, leaf-off (pilot area: late March 2014, about 5.7 points/m²); WCS in 1 km pieces	GRB buildings (GBG) and annexes (GBA)	VRBG Refgew	190 blocks of 10 km, 55.7 GB
Brussels	DHMV II (the Flemish survey covers Brussels completely)	UrbIS Buildings (249,072)	UrbIS Region	0.95 GB
Wallonia	SPW MNS and MNT 2013–2014; flown 12/2012–03/2014, about 1.5 points/m²; one GeoTIFF per province, EPSG:31370	PICC building footprints (3.87 M)	SPW administrative limits	112.9 GB of ZIPs
check only	SPW forest mask 2018 (Gembloux Agro-Bio Tech / ULiège; orthophotos 2006–2018, woody vegetation ≥ about 2 m)	–	–	30 m share

- Point clouds were not used: about 1.5–3 TB for Flanders alone. The newer Walloon survey of 2021–22 (0.5 m, about 13 points/m²) was not used for the whole region (about 400 GB) and is eight years later than DHMV II.
- Attribution required on maps and documents: *Digitaal Vlaanderen* (open data licence Flanders v1.0) and *« Sources des données : SPW »*; UrbIS is published CC-0 (to be confirmed for the buildings).

4 Methods

Figure 2 (end of this section) shows what happens to the data, step by step, on a real example; Figure 3 the whole workflow from the three regional surveys to the delivered files.

4.1 Canopy height at 1 m (encoded like the EA VOM)

- 1. Height = DSM − DTM.
- 2. Heights above 60 m → 0 (not vegetation; done first so that gap filling cannot spread them).
- 3. Building footprints → 0; NoData temporarily 0.
- 4. Gap filling: grey-scale closing with a 3 × 3 m window (a maximum then a minimum filter) fills holes up to about 2 m wide in crowns without widening crowns or forest edges.
- 5. Heights below 2.5 m → 0 (the EA VOM has no values between 0 and 2.5 m); buildings → 0 again.
- 6. NoData where the DSM, DTM or the region has no data.

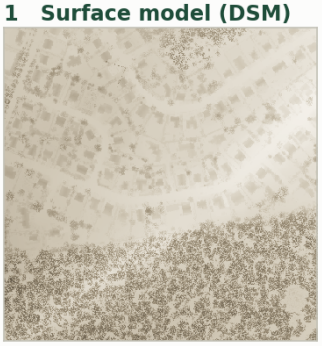
4.2 Metrics, grid and tiling

- The four metrics follow *source/misc_functions.R* of the England code (na.rm = TRUE), checked against hand-computed values in unit tests (including NoData and all-zero cells).
- Grid: 30 m cells in EPSG:31370, edges on multiples of 30 m (origin 0, 0). Chosen provisionally; any other grid can be produced from the same code (one constant).
- Processing in tiles of 6 × 6 km with a 10 m halo, so gap filling and cells at download-block edges come out as in one large raster. Checked: no seams at tile or block edges.
- Regional borders: each region's canopy height is made from its own heights and buildings inside its own border; the 1 m heights of the regions are joined before the 30 m cells are computed, so border cells are complete. Borders overlap by 1.45 ha in total; such pixels go to the first region (Flanders, Brussels, Wallonia).

From LiDAR surface models to four metrics per 30 m cell

Example: 450 × 450 m in Flemish Brabant (x 159.00–159.45 km, y 166.50–166.95 km, EPSG:31370), DHMV II (Digitaal Vlaanderen). Blue = height above ground, 0–35 m; brown = height above sea level. Every step is computed with the project code.

1 Surface model (DSM)



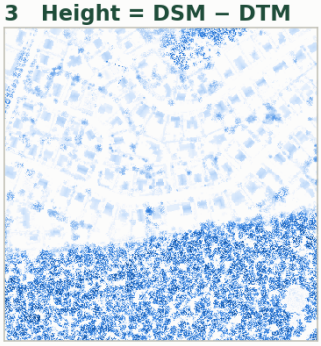
Height above sea level of the first surface the laser hits: ground, roofs, tree crowns.

2 Terrain model (DTM)



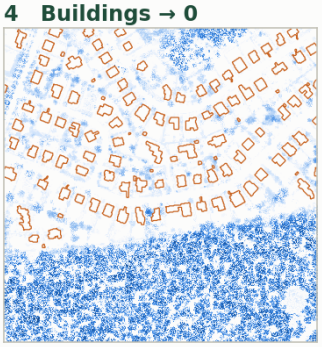
The bare ground, with buildings and vegetation removed by the data provider.

3 Height = DSM − DTM



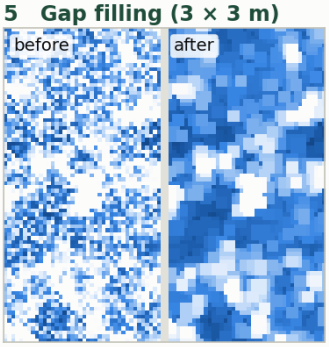
Height above ground of everything, per 1 m pixel: houses and trees alike.

4 Buildings → 0



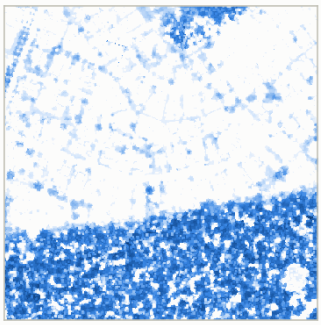
Building footprints (orange: GRB here; UrbIS, PICC elsewhere) and anything above 60 m (cranes, turbines) become 0.

5 Gap filling (3 × 3 m)



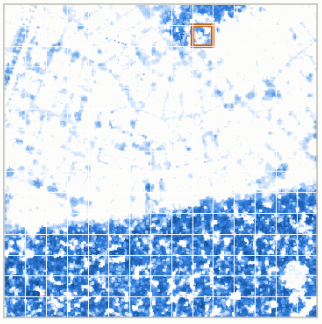
Close-up of 40 × 80 m of forest: laser hits on the ground through leaf-off crowns are filled; clearings stay.

6 Below 2.5 m → 0



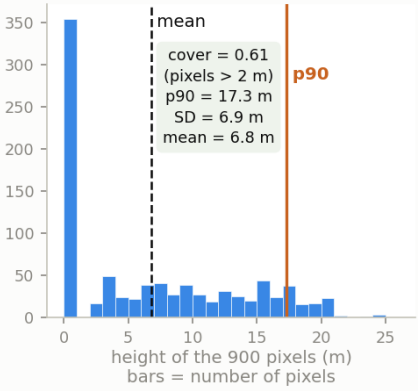
Encoded like the England VOM: vegetation of at least 2.5 m, 0 everywhere else. Zeros count in the metrics.

7 30 m cells



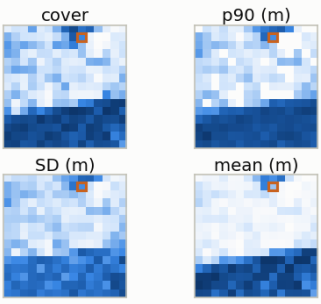
Each 30 m cell holds 900 pixels of 1 m. The orange cell is followed in step 8.

8 One cell → four metrics



The bar at 0 m holds the 39% of pixels without vegetation; they count, as in England.

9 The four layers



One value per 30 m cell: the delivered bands. The orange cell is the one of step 8.

Figure 2. The processing steps on a 450 × 450 m example (DHMV II), each computed with the project code; panel 8 shows how the 900 pixels of one 30 m cell give the four metrics.

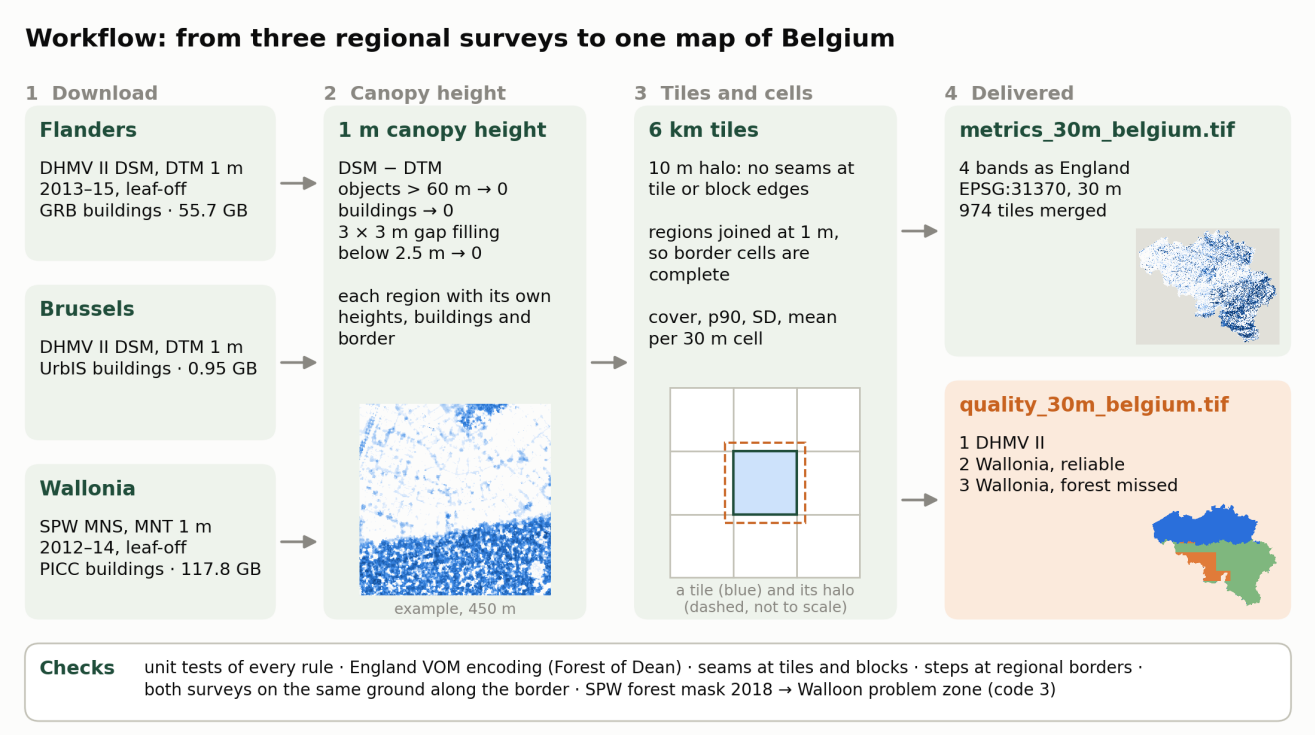


Figure 3. Workflow from the three regional surveys to the delivered files, with the checks.

5 Decisions and deviations from England

date	decision	reason and effect
7 Nov	No-vegetation pixels are 0, not NoData; zeros count in all metrics	Checked in two EA VOM tiles (Forest of Dean): no NoData at all, every other pixel ≥ 2.5 m, and the England code aggregates the zeros. Without zeros, cover would be 1 wherever there is any vegetation.
7 Nov	**Gap filling 3 m (deviation)**	Leaf-off DHMV II: in closed tall forest 40 % of pixels were ground hits through bare crowns, cover 0.60 against 0.90 in the EA VOM. With 3 m closing: 17 % and 0.86; 5 m smeared crowns (0.95). Side effect: p90 and mean in forest rise by about 3–6 m.
7 Nov	**Heights > 60 m → 0 (deviation)**	DSM − DTM contains cranes, wind turbines, masts (up to 191 m); the tallest Belgian trees are about 50 m. The EA VOM holds vegetation only, so England needed no such rule.
7 Nov	Raster DSM/DTM, not point clouds	Point clouds 1.5–3 TB for Flanders; the raster route keeps the England VOM logic.
7 Nov	Wallonia: 2013–14 survey, not 2021–22	Same period, resolution and CRS as DHMV II; 2021–22 about 400 GB. Tested on the same ground (Grez-Doiceau): per-cell correlation 0.98–0.99; Walloon cover +0.04, p90 +1.2–1.4 m (crowns slightly fuller). Offset accepted, no harmonisation.
8 Nov	Walloon flight-block problem: keep 2013–14, deliver a quality layer	See section 7. Replacing the zone with the 2021–22 survey (about 150–200 GB) remains possible later.

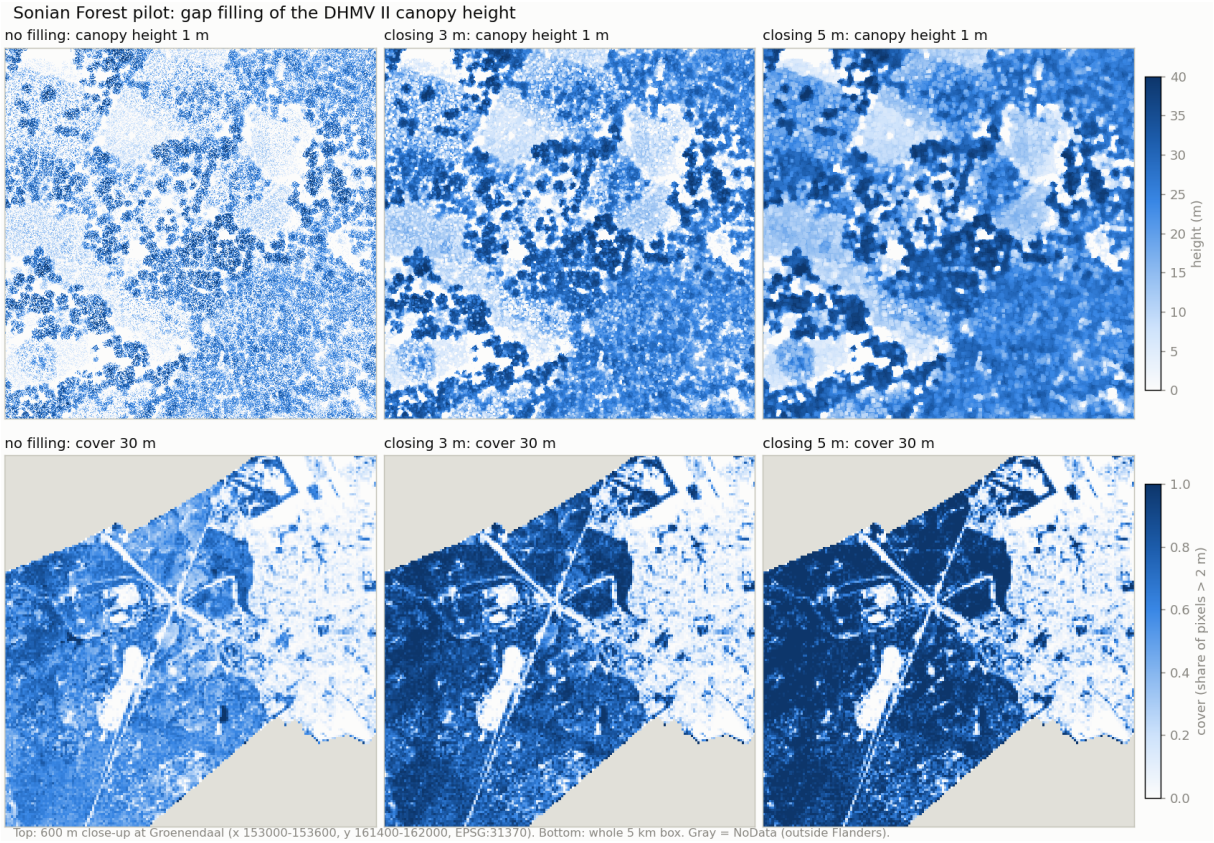


Figure 4. Sonian Forest pilot (DHMV II): canopy height at 1 m (top, 600 m close-up) and 30 m cover (bottom, 5 km) without filling, and with 3 m and 5 m closing. The 3 m closing removes the speckle of ground hits in leaf-off broadleaf stands and keeps clearings and forest lanes.

6 Results

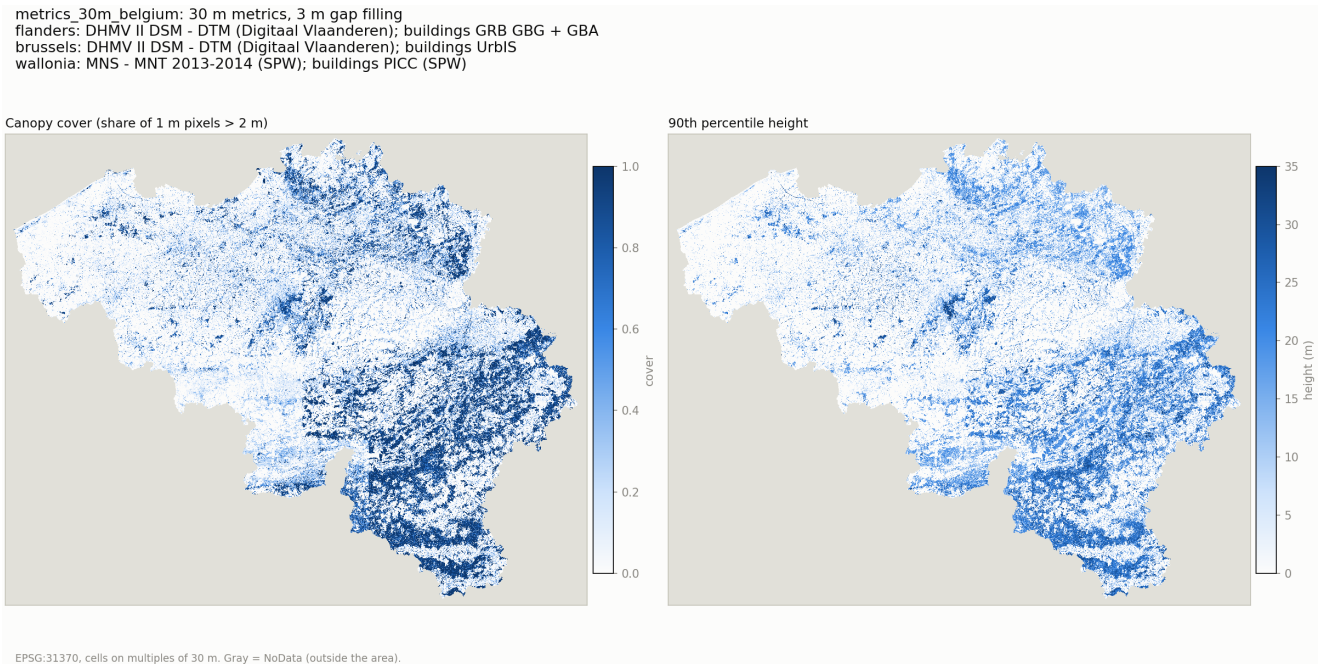


Figure 5. Canopy cover and 90th percentile height of Belgium at 30 m. The pale area in south-west Wallonia is the data-quality zone of section 7.

30 m cells	Flanders	Brussels	Wallonia	Belgium	Forest of Dean (EA VOM)
area with data	13,646 km²	164 km²	16,926 km²	30,718 km²	two 5 km tiles
forest cells (cover ≥ 0.3)	20.2 %	30.6 %	33.7 %	27.7 %	48.9 % / 85.2 %
in forest cells, median p90	17.4 m	16.8 m	21.0 m	19.6 m	20.6–21.1 m
median SD	5.5 m	6.3 m	5.5 m	5.5 m	6.7–6.9 m
median mean height	9.1 m	7.0 m	13.5 m	11.8 m	11.3–12.3 m
median cover	0.73	0.62	0.90	0.86	–

- Forest cells (20 % in Flanders) exceed the official forest area (about 11 %): 30 m cells with ≥ 30 % tree cover also include tree rows, parks and wooded gardens. Wallonia's 33.7 % matches its forest share of about a third.
- Very tall cells (p90 > 50 m) are rare: 575 in Belgium, mostly port structures of 50–60 m (Antwerp, Zeebrugge, Ghent). Pylons and overhead lines that are not in the building data stay (low cover, p90 0 m).
- SD is NaN in a few hundred border cells with a single valid pixel (as R gives NA).
- Seams: neighbouring-cell differences across 6 km tiles and 10 km download blocks equal those elsewhere (ratios 0.94–1.02).

6.1 Regional borders

A step at a regional border shows a difference between surveys. Flanders–Brussels (same survey, other building data) is the control.

forest cells, 500 m bands next to the border	cover	p90	mean height
Flanders → Brussels (control, same survey)	−0.02	−0.4 m	–
Flanders → Wallonia	−0.13	−0.9 m	−2.3 m

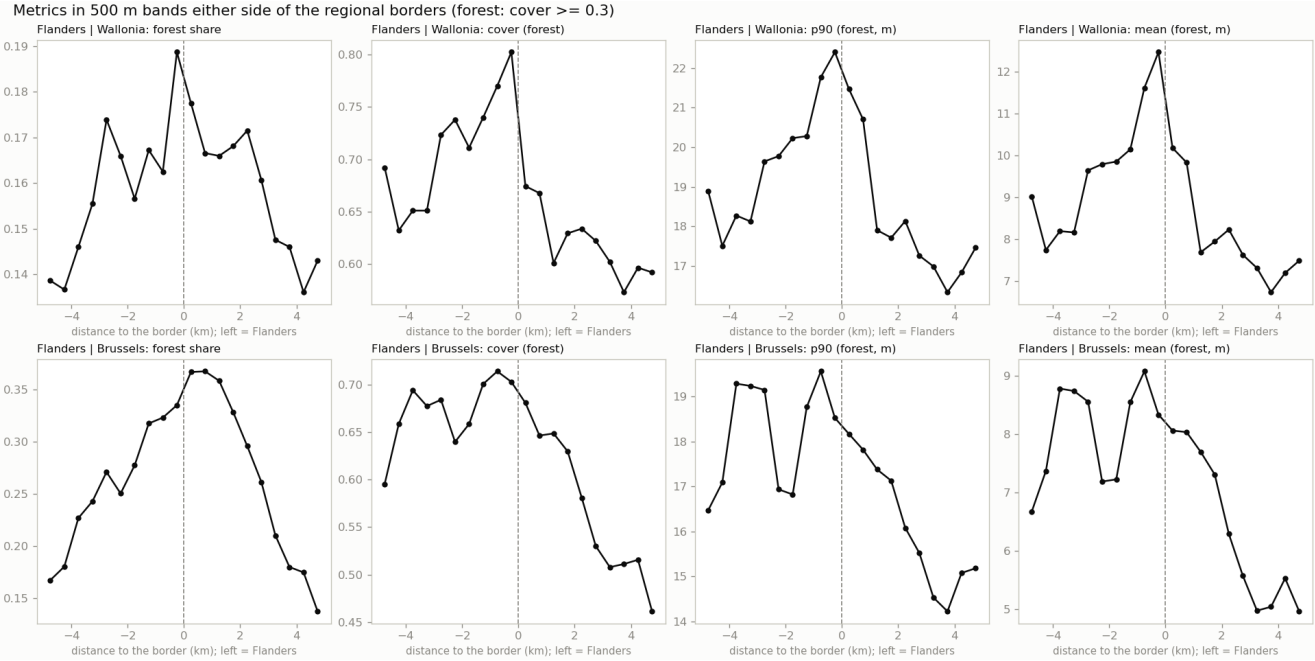


Figure 6. Metrics in 500 m bands on either side of the regional borders (left = Flanders).

7 Data quality in Wallonia: missing leaf-off broadleaf crowns

7.1 Both surveys along the Flemish-Walloon border

The Flemish survey also covers a strip of Wallonia. For the 2 km of Wallonia along the whole border both surveys were processed with identical rules and compared cell by cell (655,432 complete cells). The difference is not one offset: it changes sign along the border.

forest cells (DHMV II cover ≥ 0.3): Wallonia minus DHMV II	cover	p90	mean height
whole strip (mean)	−0.04	+0.7 m	+0.4 m
west, Hainaut (x 40–120 km)	−0.10 to −0.34	−1.9 to −5.0 m	−1.3 to −7.0 m
centre and east (x 140–260 km)	+0.04 to +0.06	+1.3 to +4.8 m	+1.3 to +4.6 m

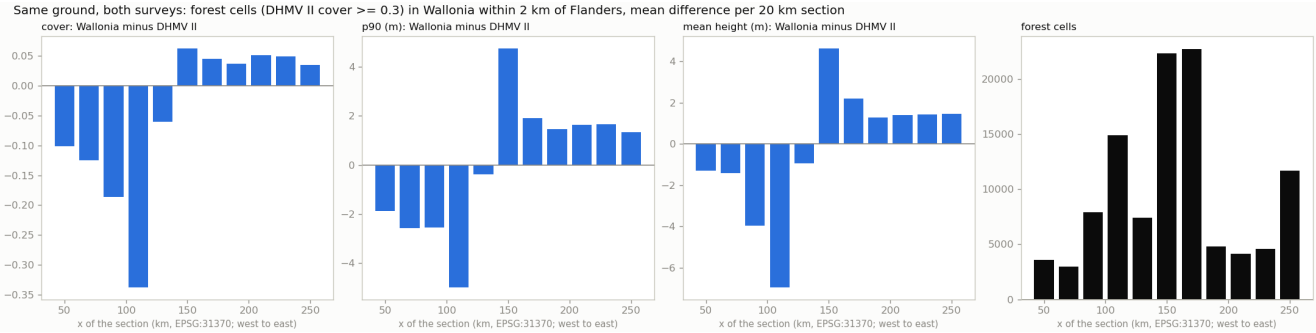


Figure 7. Same ground, both surveys: mean difference per 20 km section of the border strip, west to east.

In a 1 m close-up of broadleaf forest in the west, the Walloon DSM lies on the ground: 10 % of pixels above 2.5 m against 58 % in DHMV II, while the terrain models agree within 4 cm. Leaf-off crowns were hardly recorded by those flights; gap filling cannot repair this. Point density alone does not explain it; flight date and settings per flight block are the likely cause (SPW publishes no flight records for this survey).

7.2 Extent inside Wallonia

Inside Wallonia there is no second survey, so the SPW forest mask 2018 (from orthophotos, independent of both LiDAR surveys) served as reference: in cells that are entirely forest in the mask, a good survey finds forest (cover ≥ 0.3) in most cells. Along the border DHMV II finds 95 % of them. The result is the zone with code 3 in the quality layer (Figure 8); the evidence per 6 km tile is in Figure 9.

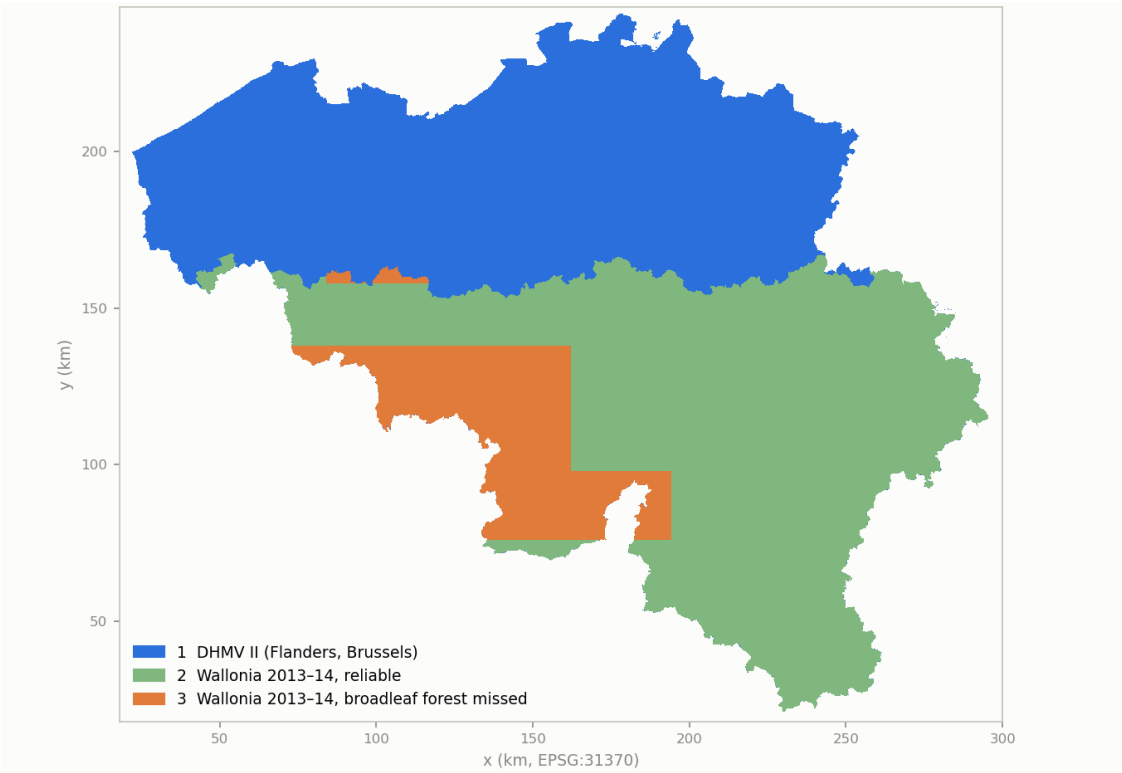


Figure 8. The quality layer (quality_30m_belgium.tif), shown at 150 m.

- The tiles fall into two groups: most find 80–100 % of the mask forest (as good as DHMV II, including the broadleaf-rich Condroz and Famenne); about 90 tiles find only 10–60 %. Few tiles lie in between, and the edges are straight lines: flight blocks.
- The zone (code 3) is three rectangles clipped to Wallonia, with edges on whole kilometres where the share jumps in 1 km profiles: x < 162 km, y 76–138 km (Hainaut south of Tournai and Ath: Mons,

Charleroi, Thuin); x 162–194 km, y 76–98 km (Entre-Sambre-et-Meuse west of the Meuse); x 84–118 km, y ≥ 158 km (around Flobecq, Lessines, Enghien).

- Check of the zone: in SPW mask forest the survey finds forest in 44.2 % of cells inside the zone and in 93.1 % in the rest of Wallonia.

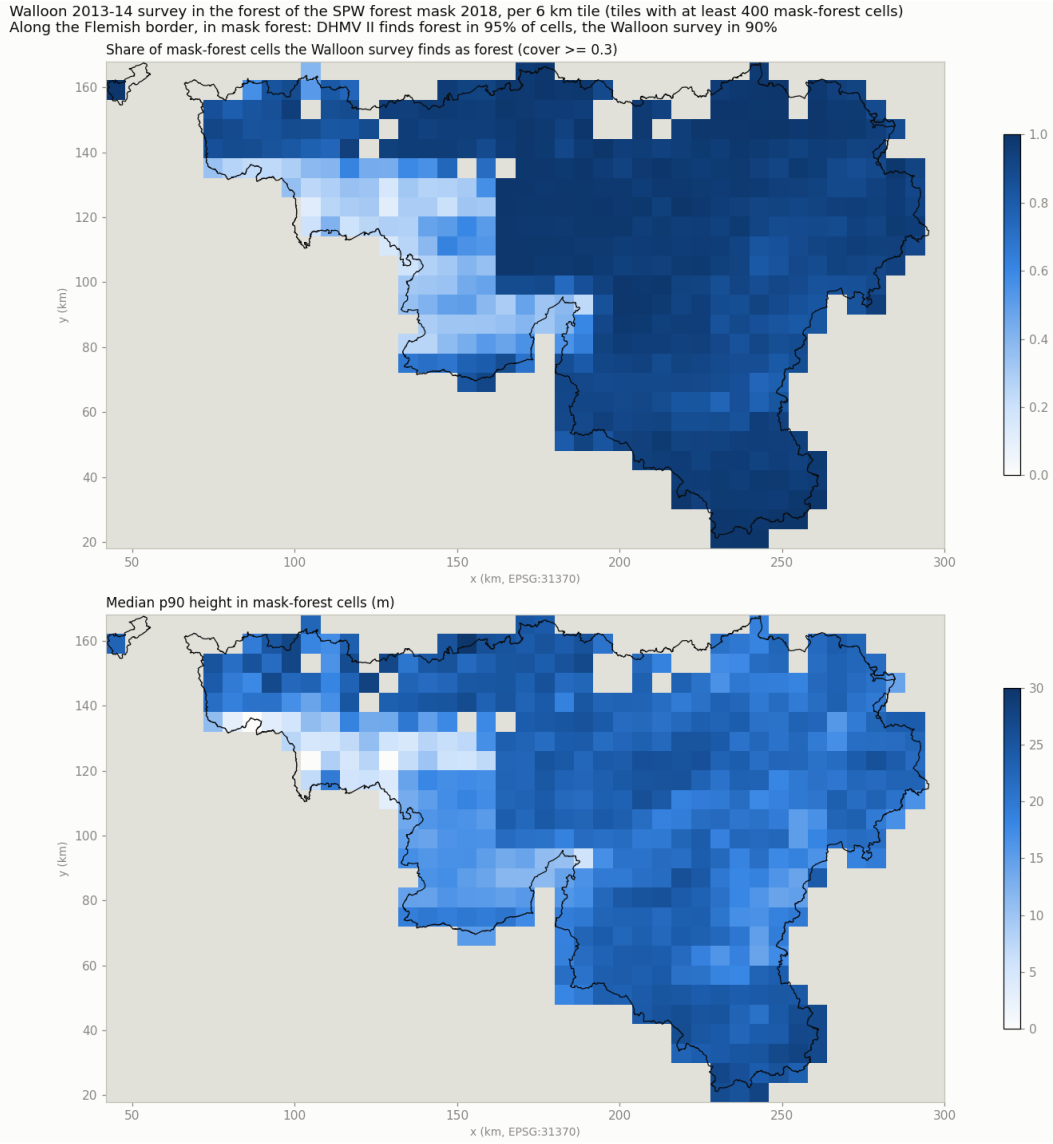


Figure 9. Share of SPW mask-forest cells that the Walloon 2013–14 survey finds as forest (top) and their median p90 (bottom), per 6 km tile.

8 Notes for the modelling

Code 3 cells (Walloon data-quality zone)
Cover, mean height and p90 are far too low in broadleaf forest there. Leave these cells out when fitting (presence and background points), and mark the zone in predictions and maps, or use the polygon to clip. Forest masks derived from cover ≥ 0.3 will also miss forest in the zone.

Heights are not exactly comparable with England
The 3 m gap filling makes cover and the 0.3 forest threshold comparable with the EA VOM, but p90 and mean in forest are about 3–6 m higher than without filling. Walloon heights are about 1 m above Flemish ones on the same ground (codes 1 and 2). A Belgian model fitted on these layers is unaffected by the first point; a transfer of the England model is affected by both.

- A transfer of the England model would also need England's PCA centring and scaling (S5 saves only the loadings).

- England reprojected the layers bilinearly (terra::project); cover and the forest threshold are sensitive to that choice. Delivering on the model's own grid avoids a second resampling.
- The surveys date from 2012–2015; forest change since then (felling, storm and bark-beetle damage, mostly in Walloon spruce) is not in the layers.

9 Open questions for the modelling lead

- Grid, CRS and extent: EPSG:3812, 31370 or 3035; Belgium only or with a border buffer; alignment to a reference grid as England's 300 m grid?
- Transfer of the England model, or a new Belgian fit?
- Is the time gap between the LiDAR (2012–2015) and the ant records a concern?
- Should we also deliver the forest mask, the distance to forest and forest-patch layers (S4b, S11)?
- Is the quality layer enough for the Walloon zone, or should it be replaced with the 2021–22 survey?

10 Repository and reproducibility

Code: github.com/dvanderzande/belgium-lidar-forest-metrics, version 0.1.0, MIT licence. Python with numpy, scipy, rasterio, geopandas and pyproj (conda environment in *environment.yml*); the working folder is set in a local *dev.env* (template *dev.env.example*). Unit tests: *python -m unittest discover tests* (15 tests, all pass). Methods, checks and decisions in detail: *docs/methods.md*; changes: *CHANGELOG.md*; how to cite: *CITATION.cff*.

part	what it does
belgium_lidar_metrics/canopy.py	canopy height rules (2.5 m, 60 m, buildings, 3 m gap filling)
belgium_lidar_metrics/metrics.py	the four England metrics per 30 m cell
belgium_lidar_metrics/download.py, output.py	WCS/WFS downloads with retries and size limit; tiled GeoTIFF writing with provenance tags
scripts/download_flanders.py, download_brussels.py, download_wallonia.py	input downloads per region
scripts/process_region.py	canopy height and metrics per 6 km tile for any regions, then the merged map
scripts/check_metrics.py, check_borders.py	value ranges, seams, figures; steps at regional borders
scripts/download_border_strip.py, compare_border_strip.py	both surveys on the same ground along the border
scripts/download_forest_mask.py, check_wallonia_quality.py	SPW forest mask and the extent of the Walloon problem
scripts/make_quality_mask.py	the quality layer and zone polygon
tests/	unit tests of the rules and metrics against hand-computed values

Run order for all of Belgium (after *source dev.env*):

```
python scripts/download_flanders.py … ; download_brussels.py … ; download_wallonia.py …
python scripts/process_region.py "$BELGIUM_LIDAR_WORK_DIR/inputs" "$BELGIUM_LIDAR_WORK_DIR/belgium"
python scripts/check_metrics.py …/belgium/metrics_30m_belgium.tif
python scripts/download_forest_mask.py …/inputs
python scripts/make_quality_mask.py …/belgium/metrics_30m_belgium.tif …/inputs
```

Run times on a laptop: downloads about 18 h in total (server-bound), processing of Belgium about 55 min (6 processes side by side). Every output GeoTIFF records its sources, rules and script in its tags. The README gives the full commands; the pilot and tests behind the decisions in section 5 are summarised in *docs/methods.md*.